Huan Li — Curriculum Vitae

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I am a postdoctoral research fellow at the Department of Computer Science, Aalborg University. Before joining Aalborg University, I was a Senior Algorithm Engineer at Ant Financial (Alipay) of Alibaba Group. I received my Ph.D. degree from Zhejiang University, China. My research interests include **databases**, **data mining**, **machine learning**, and **mobile computing**.

Positions

Postdoctoral Researcher — **Aalborg University**
Department of Computer Science — *08/05/2019–Present*
My work, under Prof. Hua Lu's research project "Data Management for Indoor LBS" funded by Independent Research Fund Denmark (DFF), is mainly engaged in managing and analyzing massive heterogeneous spatiotemporal data.

Senior Algorithm Engineer — **Ant Financial, Alibaba Group**
LBS Data Intelligence Team — *15/07/2018–05/05/2019*
My main tasks include 1) quality control and data integration of multi-source location data collected from Alibaba APPs; 2) constructing semantic models and real-time intelligent services based on user location data.

Research Assistant — **Aalborg University**
Department of Computer Science — *01/09/2014–01/02/2015*
I participated in an academic collaboration project during this period, working with Prof. Hua Lu in the field of indoor mobility data management.

Education

Doctor's Degree — **Zhejiang University, China**
Computer Science and Software Engineering — *01/09/2012–30/06/2018*
Thesis: "*Uncertain Indoor Mobility Data Mining and Analytics*"
Under the supervision of Prof. Gang Chen and Prof. Lidan Shou, I developed my ideas on designing context-dependent applications in large and complex indoor environments. I published seven papers in the top conferences and journals.

Bachelor's Degree — **Sichuan University, China**
Computer Science — *01/09/2008–30/06/2012*
I was one of the best college students in the integrative competency evaluation. My GPA score ranked 2nd among the 466 students.

Publications

- Journal Articles

- **Huan Li**, Hua Lu, Gang Chen, Ke Chen, Qinkuang Chen, Lidan Shou. Towards Translating Raw Indoor Positioning Data into Mobility Semantics. *ACM Transactions on Data Science* (TDS), 35 pages, 2020.
- **Huan Li**, Hua Lu, Lidan Shou, Gang Chen, Ke Chen. Finding Most Popular Indoor Semantic Locations Using Uncertain Mobility Data. *IEEE Trans. Knowl. Data Eng.* (TKDE), 31(11): 2108-2123, 2019.
- **Huan Li**, Hua Lu, Lidan Shou, Gang Chen, Ke Chen. In Search of Indoor Dense Regions: An Approach Using Indoor Positioning Data. *IEEE Trans. Knowl. Data Eng.* (TDKE), 30(8): 1481–1495, 2018.

- Conference Papers

- **Huan Li**, Hua Lu, Muhammad Aamir Cheema, Lidan Shou, Gang Chen. Indoor Mobility Semantics Annotation Using Coupled Conditional Markov Networks. *IEEE ICDE*, 12 pages, 2020.
- Tiantian Liu, Zijin Feng, **Huan Li**, Hua Lu, Muhammad Aamir Cheema, Hong Cheng, and Jianliang Xu. Shortest Path Queries for Indoor Venues with Temporal Variations. *IEEE ICDE*, 4 pages, 2020. (Short Paper)
- Zijin Feng, Tiantian Liu, **Huan Li**, Hua Lu, Lidan Shou, Jianliang Xu. Indoor Top-k Keyword-aware Routing Query. *IEEE ICDE*, 12 pages, 2020.
- **Huan Li**, Hua Lu, Lidan Shou, Gang Chen, Ke Chen. Finding Most Popular Indoor Semantic Locations Using Uncertain Mobility Data (extended abstract). *IEEE ICDE*, 2139-2140, 2019.

- **Huan Li**, Hua Lu, Lidan Shou, Gang Chen, Ke Chen. In Search of Indoor Dense Regions: An Approach Using Indoor Positioning Data (extended abstract). *IEEE ICDE*, 2127–2128, 2019.
- **Huan Li**, Pai Peng, Hua Lu, Lidan Shou, Ke Chen, Gang Chen. E^2C^2: Efficient and Effective Camera Calibration in Indoor Environments. *ACM UbiComp*, 9–12, 2015. (Poster)

- Demonstrations

- **Huan Li**, Feichao Shi, Hua Lu, Gang Chen, Ke Chen, Lidan Shou. TRIPS: A System for Translating Raw Indoor Positioning Data into Visual Mobility Semantics. *PVLDB*, 11(12):1918–1921, 2018.
- **Huan Li**, Hua Lu, Xin Chen, Gang Chen, Ke Chen, Lidan Shou. Vita: A Versatile Toolkit for Generating Indoor Mobility Data for Real-World Buildings. *PVLDB*, 9(13):1453–1456, 2016.

- Patents

- Lidan Shou, **Huan Li**, Ke Chen, Xinyuan Luo, Gang Chen, Dawei Jiang, Tianlei Hu. Indoor Semantic Trajectory Labeling and Complementing Method in Low-sampling Positioning Environment. CN201810778020.X, published on 11 Jan, 2019.
- Ke Chen, **Huan Li**, Lidan Shou, Xinyuan Luo, Gang Chen, Dawei Jiang, Sai Wu. An Approximate Method for Searching Check-in Users with Non-real-time Positioning Data. CN201810778007.4, published on 30 Nov, 2018.
- Gang Chen, **Huan Li**, Lidan Shou, Ke Chen, Xinyuan Luo, Sai Wu, Tianlei Hu. Indoor Continuous Flow Monitoring Method for Probabilistic Streaming Mobility Data. CN201810779777.0, published on 30 Nov, 2018.

Teaching Activities

17/01/2020–present: Supervisor of SW8 project (Spring 2020, topic: Mobile Data and Location-based Services) at Aalborg University.

01/09/2019–present: Co-supervising Bang Van Bui, Research Assistant of Prof. Hua Lu, in the topic of Wi-Fi positioning data cleansing.

01/09/2019–20/01/2020: Supervisor of SW5 project (Autumn 2019, topic: Machine Intelligence and Embedded Systems) at Aalborg University.

01/05/2019–present: Co-supervising Tiantian Liu, Ph.D. student of Prof. Hua Lu, in the topic of context-aware queries for indoor LBS.

01/03/2014–30/06/2014: Teaching assistant for undergraduate course "Operating System" (Spring 2014) at Zhejiang University.

01/09/2013–30/01/2014: Teaching assistant for postgraduate course "Advanced Database Technology" (Autumn 2013) at Zhejiang University.

Professional Services

✶ **PC member** for SIGSPATIAL ISA 2018, and MDM ALIAS 2019.

✶ **Journal referee** for TKDE (2016/2019), Journal of GIS (2019), and Journal of Mobile Information Systems (2015).

✶ **External reviewer** for VLDB (2017/2018), ICDE (2019/2020), UbiComp (2014/2015), WWW (2020), CIKM (2018/2019), DASFAA (2017/2020), SIGSPATIAL (2014/2015/2017/2019/2020), IEEE BigData (2019), and ICMR (2017).

Other Services

✻ **Volunteer leader** for VLDB 2014 and CCF NDBC 2017.

✻ Campus Ambassador of Hulu Inc. at Zhejiang University from 2012 to 2015.

✻ Monitor of the No.1 PhD Class of 2012 of Zhejiang University from 2012 to 2018.

✻ Monitor of the College's Innovation Class of Sichuan University from 2009 to 2012.

Selected Awards

✯ Outstanding Student of 41st Guards Algorithm Class of Ant Financial (2018).

✯ Candidate for Outstanding Doctoral Thesis of Zhejiang University (2018).

✯ Alibaba Big Data Algorithm Contest (2016): Spatio-temporal distribution prediction of airport passengers

(top-10/3455).

✯ Triple-A Graduate Award of Zhejiang University (2015).

✯ **Outstanding PhD Student Scholarship of Zhejiang University** (2015/2016).

✯ Outstanding Graduate Leader Award of Zhejiang University (2014).

✯ Award of Honor for Graduate of Zhejiang University (2013/2015/2016).

✯ **Creative Talents Award of Sichuan University** (2012).

✯ **China National Scholarship** (2009/2010/2011).

Grants

✯ VLDB Travel Grant (2016).

✯ **EU Marie Skłodowska-Curie Individual Fellowships** (2020), 219,000 EUR.

Selected Projects

iLocation: Platform for Location-based Service at Ant Financial **Ant Financial (Alibaba)**
Project at LBS Data Intelligence Team *15/07/2018–05/05/2019*
My contributions to this product are twofold. (1) I applied machine learning algorithms to constructing location-dependent portraits (e.g., residence city, home, working place and frequented POIs) for universe users who can be either inside or outside Alipay economic system. The construction model involves hundreds of billions of input data records daily and has served many important businesses of Ant Financial. (2) I designed modules for automatic extraction, monitoring, and enhancement of ultra-large-scale location data on the bottom of the iLocation platform, leading unified management and export of real-time and offline data services.

Retail Industry Data Analysis Platform **Zhejiang University**
Project at Database Laboratory *01/03/2015–31/12/2017*
I am the CO-PI of this project. My main work is to explore the mobility data obtained from wireless positioning infrastructure to recognize the customer behaviors in a shopping mall. Our recognized behaviors in the form of semantic trajectories are beneficial to customer marketing and mall management. One demonstration has been showcased in PVLDB and one full paper has been accepted by ACM TDS. The detail information can be found at `https://longaspire.github.io/trips`.

Management and Mining of Complex Data in Intelligent Indoor Environment **Zhejiang University**
NSFC Grant No. 61672455 *01/07/2015–01/05/2016*
In the project, I designed and implemented a generic, user-configurable toolkit for generating different types of indoor mobility data for real-world buildings. It can be used to simulate high volumes of indoor mobility data to evaluate location-dependent queries/algorithms. It can also provide the additional "ground truth" for effectiveness studies in analytics. A relevant paper has been published in VLDB'16. The detail information is available at `https://longaspire.github.io/vita`.

Management and Application of Indoor Trajectory Data. **Aalborg University**
NSFC Grant No. 61528207 (Joint Research Fund for Overseas) *01/09/2014–28/02/2018*
During the period, I was a visiting researcher (advised by Prof. Hua Lu) in the Center for Data-intensive Systems (Daisy) at Aalborg University. My research at Daisy focused on designing algorithms for finding the densest regions or most popular locations with uncertain indoor mobility data. Such data mining techniques are practically useful in human's daily life (e.g., congestion control and public resource management). Two relevant papers have been published in IEEE TKDE. Shortened versions of both works have been presented as posters at ICDE'19.

Human-Computer Fusion based Crowd Cognition: Theory and Models **Zhejiang University**
973 Program of China, Grant No. 2015CB352402 *01/12/2013–30/06/2015*
In this project, my work focused on the design of a vision-enhanced localization system for mobile users. To make full use of the built-in sensor readings on smartphones, I devised a novel metric learning algorithm to fuse Wi-Fi, gyroscope, and image data feature for inferring the underlying device position and pose. A short paper has been published in UbiComp'15.